CLASS Mission Analysis Report

Cognitive Load Analytics System for Long-Duration Space Missions

Post-Mission Analytical Assessment

1. Mission Information

Parameter	Value
System	CLASS
Number of Astronauts	3
Astronaut IDs	A01, A02, A03
Mission Days	1 - 30
Total Records	30

2. Daily Analytical Results

Day	Astronaut	CLI	FRS	PSI	DES	MRS	Risk
1	A01	10.29	95.79	100.00	77.51	90.24	Low Risk
2	A01	19.22	88.88	92.12	75.14	83.66	Low Risk
3	A01	32.01	81.97	55.61	60.11	67.04	Low Risk
4	A01	43.83	72.96	52.89	56.16	59.71	Low Risk
5	A01	59.30	61.84	44.46	37.96	46.05	Moderate Risk
6	A01	76.50	50.72	14.44	19.77	27.56	High Risk
7	A01	95.11	39.61	14.44	1.58	14.65	High Risk
8	A01	84.00	52.37	52.60	18.19	32.96	High Risk
9	A01	63.25	63.49	18.13	36.39	39.62	High Risk
10	A01	46.64	73.55	16.29	53.79	51.10	Moderate Risk
11	A02	4.17	97.89	100.00	91.75	96.16	Low Risk
12	A02	18.39	89.93	47.38	75.93	75.42	Low Risk
13	A02	29.50	80.92	50.79	71.98	69.53	Low Risk
14	A02	43.99	71.91	48.68	55.37	58.36	Low Risk
15	A02	59.99	60.79	18.13	37.18	40.12	High Risk
16	A02	77.20	49.67	14.44	18.98	26.89	High Risk
17	A02	98.00	37.50	12.58	0.00	12.49	High Risk
18	A02	80.56	51.32	50.46	17.40	33.10	High Risk
19	A02	63.35	62.43	16.29	35.60	38.76	High Risk

Day	Astronaut	CLI	FRS	PSI	DES	MRS	Risk
20	A02	45.35	74.61	12.58	54.58	51.21	Moderate Risk
21	A03	0.00	100.00	100.00	100.00	100.00	Low Risk
22	A03	16.96	90.99	44.75	83.39	77.46	Low Risk
23	A03	27.68	83.03	37.15	73.56	68.27	Low Risk
24	A03	42.50	74.01	53.51	63.61	62.36	Low Risk
25	A03	55.30	62.89	47.85	51.42	51.56	Moderate Risk
26	A03	72.31	51.78	28.09	33.23	35.18	High Risk
27	A03	96.02	40.66	0.00	9.04	13.62	High Risk
28	A03	78.37	53.42	41.80	25.65	34.62	High Risk
29	A03	60.46	64.54	18.13	43.84	42.58	Moderate Risk
30	A03	42.55	75.66	14.44	62.04	54.55	Moderate Risk

3. Overall Mission Summary

Metric	Average	Minimum	Maximum
Cognitive Load Index	51.43	0.00	98.00
Fatigue Recovery Score	68.50	37.50	100.00
Performance Stability Index	40.60	0.00	100.00
Decision Efficiency Score	48.04	0.00	100.00
Mission Readiness Score	51.83	12.49	100.00

4. Risk Analysis

Risk Level	Number of Records
Low Risk	12
High Risk	12
Moderate Risk	6

5. Important Mission Events

Mission Day	Astronaut	Risk Level
6	A01	High Risk
7	A01	High Risk
8	A01	High Risk
9	A01	High Risk
15	A02	High Risk
16	A02	High Risk
17	A02	High Risk
18	A02	High Risk
19	A02	High Risk
26	A03	High Risk
27	A03	High Risk
28	A03	High Risk

6. Analytical Interpretation

CLASS integrates cognitive load, fatigue recovery, performance stability, decision efficiency, and mission readiness indicators into a unified analytical report. The results provide a structured view of astronaut-related mission conditions across the recorded mission days.

The generated indicators are intended as prototype decision-support measures based on the constructed demonstration dataset. They are not medical diagnoses or operational certification measures.

7. Analyst Notes

Notes / Observations